

K-Nearest Neighbors Algorithm

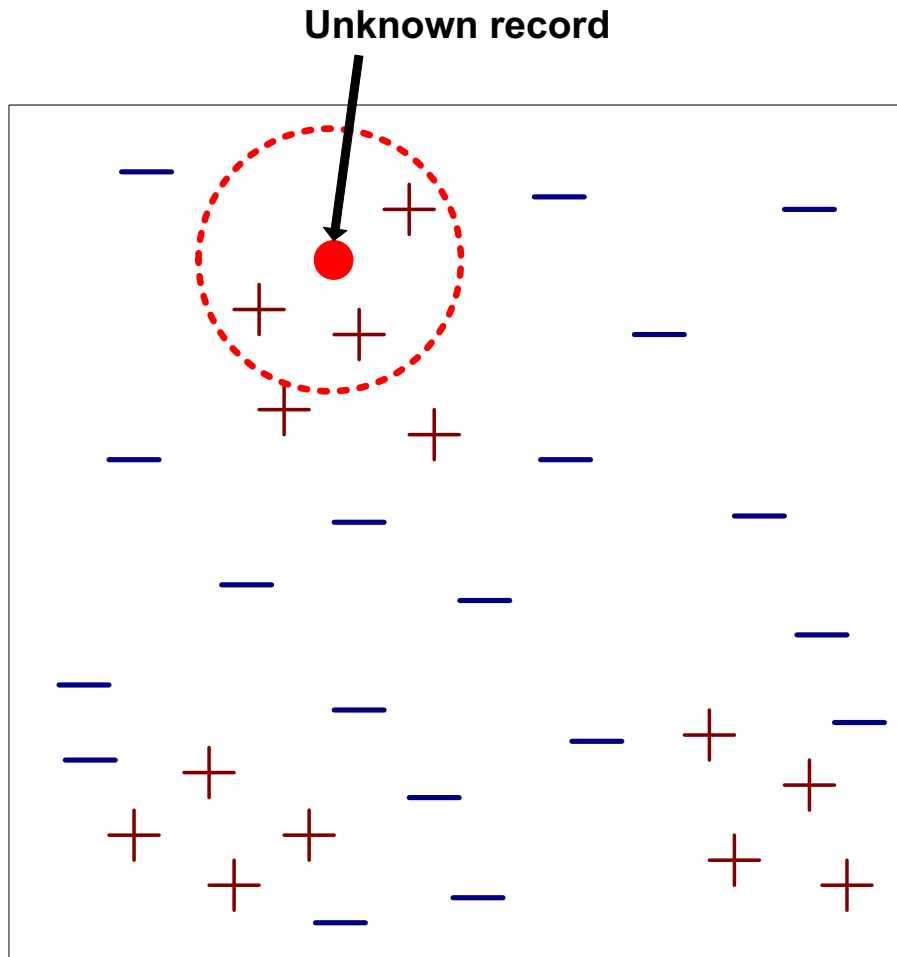
Instance Based Classifiers

- **Instance-based Classifiers** do not create a model but use training examples directly to classify unseen examples (“*lazy*” classifiers).
- **Examples**
 - **Rote-learner**
 - Memorizes entire training data and performs classification only if attributes of record match one of the training examples exactly
 - **Nearest neighbor**
 - Uses k “closest” points (nearest neighbors) for performing classification

K-Nearest Neighbor

- All instances correspond to points in an n -dimensional Euclidean space
- Classification is delayed till a new instance arrives
- Classification done by comparing feature vectors of the different points
- Target function may be discrete or real-valued

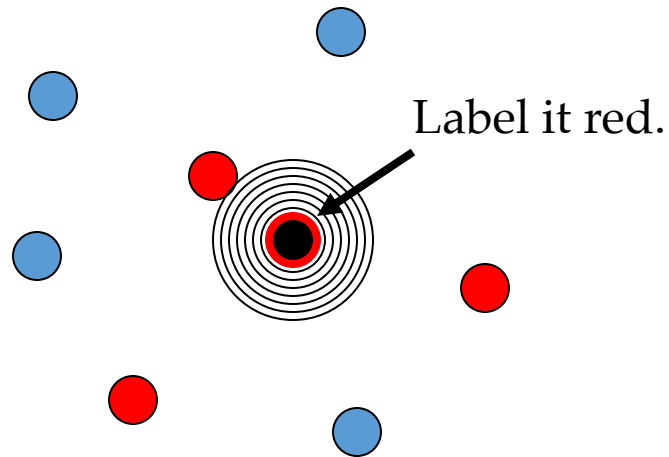
kNN— k Nearest-Neighbor Classifiers



- **Requires three things**
 - The set of stored records
 - Distance Metric to compute distance between records
 - The value of k , the number of nearest neighbors to retrieve
- **To classify an unknown record:**
 - Compute distance to other training records
 - Identify k nearest neighbors
 - Use class labels of nearest neighbors to determine the class label of unknown record (e.g., by taking majority vote)

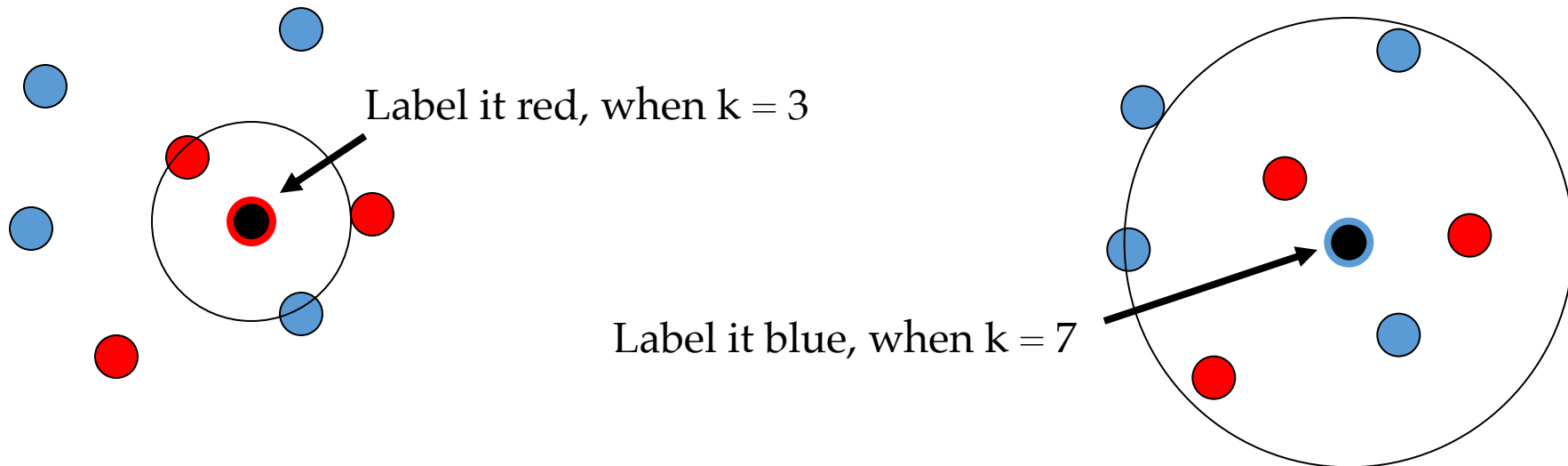
1-Nearest Neighbor

- One of the simplest of all machine learning classifiers
- Simple idea: label a new point the same as the closest known point



k – Nearest Neighbor

- Generalizes 1-NN to smooth away noise in the labels
- A new point is now assigned **the most frequent label of its k nearest neighbors**



K-Nearest Neighbor Classification

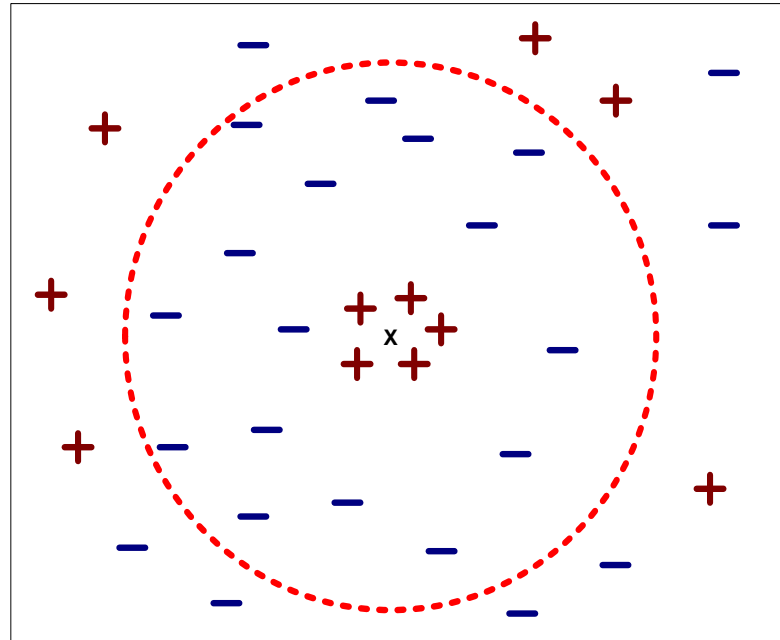
- **Compute distance between two points (Euclidean distance)**

$$d(p, q) = \sqrt{\sum_i (p_i - q_i)^2}$$

- **Determine the class from nearest neighbor list**
 - Take the majority vote of class labels among the k-nearest neighbors
 - Weigh the vote according to distance
 - weight factor, $w = 1/d^2$

K-Nearest Neighbor Classification

- **Choosing the value of k**
 - If k is too small, sensitive to noise points
 - If k is too large, neighborhood may include points from other classes



K-Nearest Neighbor Classification

- **Step #1** - Assign a value to **K**.
- **Step #2** - **Calculate the distance** between the new data entry and all other existing data entries. **Arrange them in ascending order.**
- **Step #3** - Find the **K** nearest neighbors to the new entry based on the calculated distances.
- **Step #4** - Assign the new data entry to the majority class in the nearest neighbors.

Brightness	Saturation	Class
40	20	Red
50	50	Blue
60	90	Blue
10	25	Red
70	70	Blue
60	10	Red
25	80	Blue

let's assume the value of K is 5.

Brightness	Saturation	Class
20	35	?

To calculate the distance from the new entry to other entries in the data set using the Euclidean distance formula.

$$\sqrt{(X_2 - X_1)^2 + (Y_2 - Y_1)^2}$$

Where:

X_2 = New entry's brightness (20).

X_1 = Existing entry's brightness.

Y_2 = New entry's saturation (35).

Y_1 = Existing entry's saturation.

Brightness	Saturation	Class
40	20	Red
50	50	Blue
60	90	Blue
10	25	Red
70	70	Blue
60	10	Red
25	80	Blue

Distance #1

For the first row, d1:

Brightness	Saturation	Class
20	35	?

$$\begin{aligned}
 d1 &= \sqrt{(20 - 40)^2 + (35 - 20)^2} \\
 &= \sqrt{400 + 225} \\
 &= \sqrt{625} \\
 &= 25
 \end{aligned}$$

Brightness	Saturation	Class	Distance
10	25	Red	10
40	20	Red	25
50	50	Blue	33.54
25	80	Blue	45
60	10	Red	47.17
70	70	Blue	61.03
60	90	Blue	68.01

We chose 5 as the value of K, we'll only consider the first five rows.

Brightness	Saturation	Class	Distance
10	25	Red	10
40	20	Red	25
50	50	Blue	33.54
25	80	Blue	45
60	10	Red	47.17

The **majority class within the 5 nearest neighbors to the new entry is Red**. Therefore, we'll classify the **new entry as Red**.

Apply K nearest neighbor classifier to predict the diabetic patient with the given features BMI, Age. If the training examples are,

BMI	Age	Sugar
33.6	50	1
26.6	30	0
23.4	40	0
43.1	67	0
35.3	23	1
35.9	67	1
36.7	45	1
25.7	46	0
23.3	29	0
31	56	1

BMI	Age	Sugar	Formula	Distance
33.6	50	1	$\sqrt{((43.6-33.6)^2+(40-50)^2)}$	14.14
26.6	30	0	$\sqrt{((43.6-26.6)^2+(40-30)^2)}$	19.72
23.4	40	0	$\sqrt{((43.6-23.4)^2+(40-40)^2)}$	20.20
43.1	67	0	$\sqrt{((43.6-43.1)^2+(40-67)^2)}$	27.00
35.3	23	1	$\sqrt{((43.6-35.3)^2+(40-23)^2)}$	18.92
35.9	67	1	$\sqrt{((43.6-35.9)^2+(40-67)^2)}$	28.08
36.7	45	1	$\sqrt{((43.6-36.7)^2+(40-45)^2)}$	8.52
25.7	46	0	$\sqrt{((43.6-25.7)^2+(40-46)^2)}$	18.88
23.3	29	0	$\sqrt{((43.6-23.3)^2+(40-29)^2)}$	23.09
31	56	1	$\sqrt{((43.6-31)^2+(40-56)^2)}$	20.37

BMI	Age	Sugar	Distance	Rank
33.6	50	1	14.14	2
26.6	30	0	19.72	
23.4	40	0	20.20	
43.1	67	0	27.00	
35.3	23	1	18.92	
35.9	67	1	28.08	
36.7	45	1	8.52	1
25.7	46	0	18.88	3
23.3	29	0	23.09	
31	56	1	20.37	

Now, we need to apply the majority voting technique to decide the resulting label from the new example. Here the 1st and 2nd nearest neighbors have target label 1 and the 3rd nearest neighbor has target label 0. Target label 1 has the majority. Hence the new example is classified as 1, That is the diabetic patient has Sugar

KNN Regression

- In KNN regression, we predict the value of a dependent variable for a data point based on the average or mean of the target values of its K nearest neighbors in the training data.
- The value of K is a user-defined hyperparameter. It determines the number of nearest neighbors used to make the prediction.

KNN Regression Algorithm

- **Choose a value for K:** We first choose a value for K. This determines the number of nearest neighbors used to make the prediction.
- **Calculate the distance:** After choosing K, we calculate the distance between each data point in the training set and the target data point for which a prediction is being made. For this, we can use a variety of distance metrics, including Euclidean distance, Manhattan distance, or Minkowski distance.
- **Find the K nearest neighbors:** After calculating the distance between the existing data points and the new data point, we identify K nearest neighbors by selecting the K data points nearest to the new data point.
- **Calculate the prediction:** After finding the neighbors, we calculate the value of the dependent variable for the new data point. For this, we take the **average of the target values of the K nearest neighbors**.

Length	Weight	Cost
10	15	45
11	6	37
12	14	48
7	9	33
9	14	38
8	12	40
6	11	35
15	10	50
14	8	46
7	12	35
10	6	36
13	8	44
9	7	32
5	8	30
5	10	30

- The dataset contains the length and weight of metal rods along with their cost.
- Now, suppose that we want to calculate the cost for a rod with a length of 7 and a weight of 8.

Point	Distance from (7,8)
(10, 15)	7.61
(11, 6)	4.47
(12, 14)	7.81
(7, 9)	1.0
(9, 14)	6.32
(8, 12)	4.12
(6, 11)	3.16
(15, 10)	8.24
(14, 8)	7.0
(7, 12)	4.0
(10, 6)	3.60
(13, 8)	6.0
(9, 7)	2.23
(5, 8)	2.0
(5, 10)	2.82

Find the three closest points in the dataset. For this, we will sort the points according to their distances from (7, 8).

Point	Distance from (7,8)
(7, 9)	1
(5, 8)	2
(9, 7)	2.23
(5, 10)	2.82
(6, 11)	3.16
(10, 6)	3.6
(7, 12)	4
(8, 12)	4.12
(11, 6)	4.47
(13, 8)	6
(9, 14)	6.32
(14, 8)	7
(10, 15)	7.61
(12, 14)	7.81
(15, 10)	8.24

you can observe that the three points closest to (7, 8) are (7, 9), (5, 8), and (9, 7). These points have costs of 33, 30, and 32.

To calculate the cost of a rod with length 7 and weight 8, we can take the average of the above costs. Hence, the cost for (7, 8) will be 31.67.